

NAME

RDKitEnumerateStereoisomers.py - Enumerate stereoisomers of molecules

SYNOPSIS

```
RDKitEnumerateStereoisomers.py [--discardNonPhysical <yes or no>] [--infileParams <Name,Value,...>] [
--mode <UnassignedOnly or All>] [--maxIsomers <number>] [--outfileParams <Name,Value,...>] [
--overwrite] [-w <dir>] -i <infile> -o <outfile>
```

```
RDKitEnumerateStereoisomers.py -h | --help | -e | --examples
```

DESCRIPTION

Perform a combinatorial enumeration of stereoisomers for molecules around all or unassigned chiral atoms and bonds.

The supported input file formats are: Mol (.mol), SD (.sdf, .sd), SMILES (.smi, .csv, .tsv, .txt)

The supported output file format are: SD (.sdf, .sd), SMILES (.smi)

OPTIONS

-d, --discardNonPhysical <yes or no> [default: yes]

Discard stereoisomers with non-physical structures. Possible values: yes or no. The non-physical nature of a stereoisomer is determined by embedding the structure to generate a conformation for the stereoisomer using standard distance geometry methodology.

A word to the wise from RDKit documentation: this is computationally expensive and uses a heuristic that could result in loss of stereoisomers.

-e, --examples

Print examples.

-m, --mode <UnassignedOnly or All> [default: UnassignedOnly]

Enumerate unassigned or all chiral centers. The chiral atoms and bonds with defined stereochemistry are preserved.

--maxIsomers <number> [default: 50]

Maximum number of stereoisomers to generate for each molecule. A value of zero indicates generation of all possible stereoisomers.

-h, --help

Print this help message.

-i, --infile <infile>

Input file name.

--infileParams <Name,Value,...> [default: auto]

A comma delimited list of parameter name and value pairs for reading molecules from files. The supported parameter names for different file formats, along with their default values, are shown below:

```
SD, MOL: removeHydrogens,yes,sanitize,yes,strictParsing,yes
SMILES: smilesColumn,1,smilesNameColumn,2,smilesDelimiter,space,
        smilesTitleLine,auto,sanitize,yes
```

Possible values for smilesDelimiter: space, comma or tab.

-o, --outfile <outfile>

Output file name.

--outfileParams <Name,Value,...> [default: auto]

A comma delimited list of parameter name and value pairs for writing molecules to files. The supported parameter names for different file formats, along with their default values, are shown below:

```
SD: compute2DCoords,auto,kekulize,yes,forceV3000,no
SMILES: smilesKekulize,no,smilesDelimiter,space, smilesIsomeric,yes,
```

smilesTitleLine,yes

Default value for compute2DCoords: yes for SMILES input file; no for all other file types.

--overwrite

Overwrite existing files.

-w, --workingdir <dir>

Location of working directory which defaults to the current directory.

EXAMPLES

To enumerate only unassigned atom and bond chiral centers along with discarding of non-physical structures, keeping a maximum of 50 stereoisomers for each molecule, and write out a SMILES file, type:

```
% RDKitEnumerateStereoisomers.py -i Sample.smi -o SampleOut.smi
```

To enumerate only unassigned atom and bond chiral centers along with discarding any non-physical structures, keeping a maximum of 250 stereoisomers for a molecule, and write out a SD file, type:

```
% RDKitEnumerateStereoisomers.py --maxIsomers 0 -i Sample.smi  
--maxIsomers 250 -o SampleOut.sdf
```

To enumerate all possible assigned and unassigned atom and bond chiral centers, without discarding any non-physical structures, keeping a maximum of 500 stereoisomers for a molecule, and write out a SD file, type:

```
% RDKitEnumerateStereoisomers.py -d no -m all --maxIsomers 500  
-i Sample.smi -o SampleOut.sdf
```

To enumerate only unassigned atom and bond chiral centers along with discarding of non-physical structures, keeping a maximum of 50 stereoisomers for each molecule in a CSV SMILES file, SMILES strings in column 1, name in column 2, and write out a SD file without kekulization, type:

```
% RDKitEnumerateStereoisomers.py --infileParams  
"smilesDelimiter,comma,smilesTitleLine,yes,smilesColumn,1,  
smilesNameColumn,2" --outfileParams "compute2DCoords,yes,  
kekulize,no" -i SampleSMILES.csv -o SampleOut.sdf
```

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SEE ALSO

RDKitConvertFileFormat.py, RDKitEnumerateCompoundLibrary.py, RDKitGenerateConformers.py,
RDKitGenerateMolecularFrameworks.py

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The functionality available in this script is implemented using RDKit, an open source toolkit for cheminformatics developed by Greg Landrum.

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